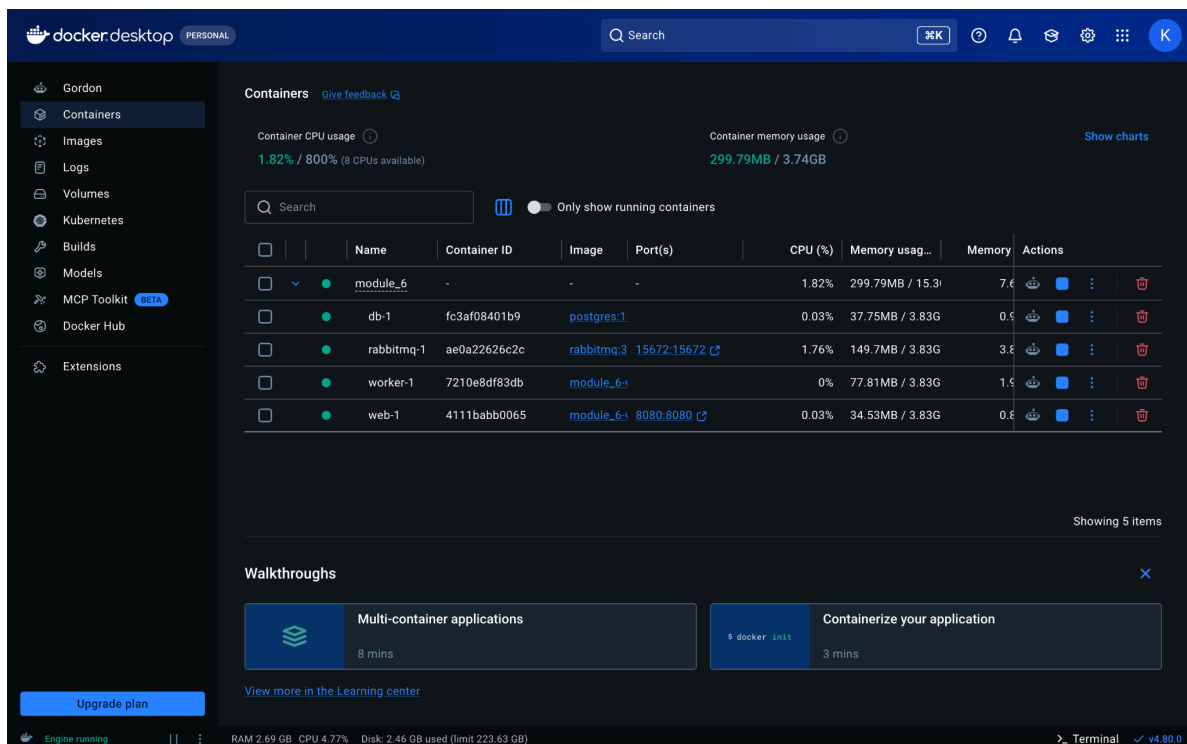


Kishore Narayanan
knaraya8@jh.edu

Module 6 - Deploy Anywhere

This project deploys the Grad Cafe Analytics application using Docker Compose. The application consists of four services: a Flask web application, PostgreSQL database, RabbitMQ message broker, and worker service. User requests are published to RabbitMQ and processed asynchronously by the worker before updating PostgreSQL.



Docker Compose Architecture:

- web: Flask application providing Grad Cafe Analytics user interface
- worker: Processes RabbitMQ tasks, performs analysis updates, updates PostgreSQL
- db: PostgreSQL database responsible for storing applicant records, watermark information, analysis data
- rabbitmq: Message broker queuing background tasks between web app and the worker

RabbitMQ

Message flow:

When a user presses the 'pull data' or 'update analysis' buttons, the application publishes a message to RabbitMQ. The worker (listening to the queue) retrieves and performs the necessary task. They then commit and acknowledge the transaction, and wait for the next task.

RabbitMQ manages background tasks between the web application and worker service. Long-running operations can execute asynchronously without blocking the web interface.

RabbitMQ 3.13.7 Erlang 26.2.5.16

Refreshed 2026-06-30 00:37:58 | Refresh every 5 seconds

Virtual host: All

Cluster: rabbit@ae0a22626c2c

User: guest | Log out

Queues

▼ All queues (1)

Page 1 of 1 - Filter: ☐ Regex

Displaying 1 item, page size up to: 100

Overview		Messages				Message rates				
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	Incoming	deliver / get	ack
/	tasks_q	Classic	0	running	0	0	0	0.00/s	0.00/s	0.00/s

➤ Add a new queue

HTTP API | Documentation | Tutorials | New releases | Commercial edition | Commercial support | Discussions | Discord | Plugins | GitHub

Database Initialization:

The PostgreSQL database is automatically initialized by Docker Compose when the stack starts. Records are loaded in incrementally using watermark tracking, and idempotent database operations help avoid duplicate records.

Worker Behavior:

The worker processes messages one at a time, helping ensure reliable execution of the task and consistent updates to the database.

Running the Application

← → ↻ localhost:8080 ☆ 📄 🗨️ 🌐 Relaunch to update

Grad Cafe Data Analysis

This page displays analysis results from the PostgreSQL applicants database.

Pull Data

Pull Data loads applicant records from the Module 2 data file into PostgreSQL. Existing records are preserved and duplicate records are skipped. This can take several minutes.

Update Analysis

Update Analysis refreshes this page with the most current query results from the PostgreSQL database.

Q1
How many entries are Fall 2026 entries?
Answer: 29608

Q2
What percentage of entries are from international students?
Answer: 45.50%

Q3
What is the average GPA, GRE, GRE V, and GRE AW?
Answer: 3.78, 165.87, 160.75, 4.36

Q4
What is the average GPA of American students in Fall 2026?
Answer: 3.79

Q5

← → ↻ localhost:8080 ☆ 📄 🗨️ 🌐 Relaunch to update

Q5
What percent of Fall 2026 entries are acceptances?
Answer: 36.94%

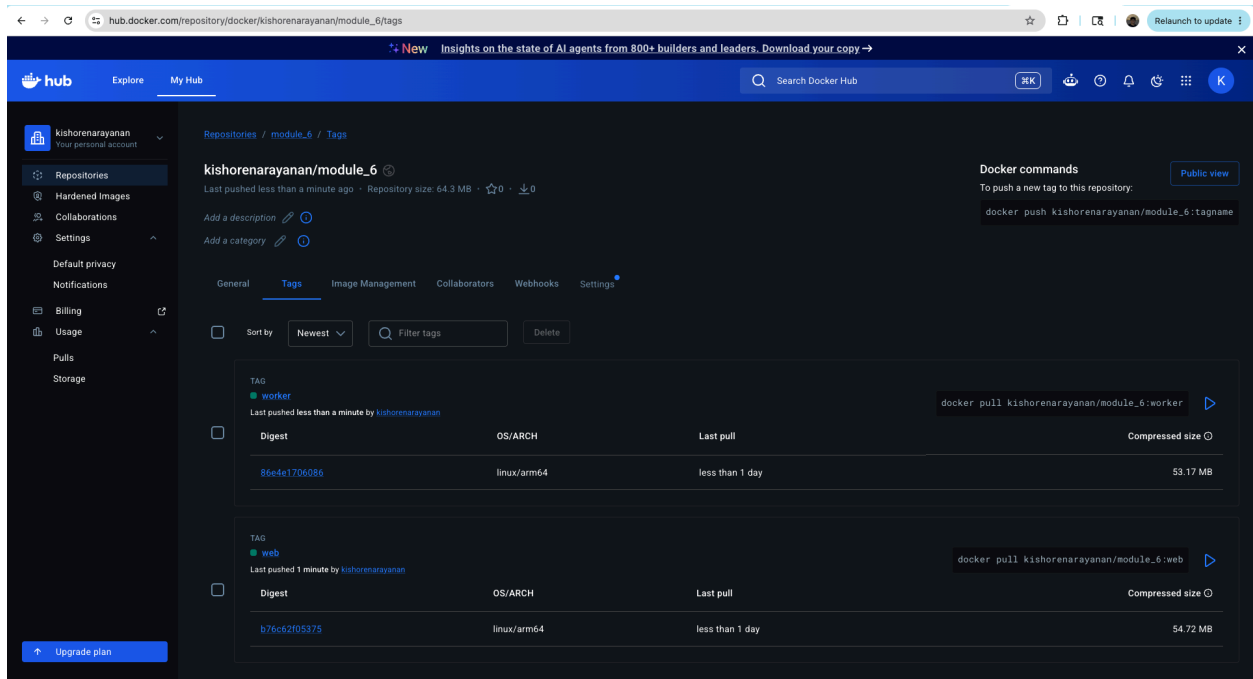
Q6
What is the average GPA of Fall 2026 accepted applicants?
Answer: 3.76

Q7
How many entries are from applicants who applied to JHU for a masters degree in Computer Science?
Answer: 8

Q8
How many 2026 PhD Computer Science acceptances are from Georgetown, MIT, Stanford, or Carnegie Mellon?
Answer: 28

Q9
Does the answer for Q8 change when using LLM generated fields?
Answer: 28, 28, No, the answer does not change.

Q10
Additional: What is the acceptance rate by degree type?
Answer: Master: 3.55%, PhD: 3.43%, Professional: 28.00%



Both Docker images (web and worker) were successfully published to Docker Hub under the module_6 repository.

Repository link: https://hub.docker.com/r/kishorenarayanan/module_6

Verification Steps

- Starting the Docker Compose stack successfully
- Accessing Flask web application
- Verifying RabbitMQ Management UI
- Passing unit tests with 100% coverage
- Achieving 10/10 Pylint score
- Completing GitHub Actions CI workflow
- Publishing Docker images to Docker Hub